

PARTS LIST

Semiconductors:

- IC1 - ATmega16, 8-bit MCU
- IC2 - 7805, 5V regulator
- LED1 - 5mm green LED
- LED2, LED3 - 5mm red LED
- T1-T3 - PN2222A npn transistor
- D1 - 1N4007 rectifier diode
- IRTX1 - IR transmitter (LED)
- IRRX1 - IR receiver (photodiode)
- LCD1 - 16x2 alphanumeric display

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R6 - 470-ohm
- R7, R8, R10 - 10-kilo-ohm
- R9 - 220-ohm
- R11 - 100-ohm
- RNW1 - 10-kilo-ohm resistor network
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1 - 10 μ F, 25V electrolytic
- C2, C3 - 0.1 μ F ceramic disk
- C4, C5 - 22pF ceramic disk

Miscellaneous:

- X_{osc}1 - 8MHz crystal oscillator
- S1-S5 - Tactile switch
- S6 - On/off switch
- CON1, CON2 - 2-pin connector terminal
- PZ1 - DC piezo buzzer
- RL1 - 12V, 1C/O relay
- JACK1 - 12V DC jack
- 12V DC power supply

sole, followed by pressing Start. The products moving on a conveyor belt are counted by the programmable object counter. The conveyor belt automatically stops when the number of counts entered by the operator is reached.

Main features of this programmable object counter are simple user interface (UI) using only four pushbuttons, an LCD display for the UI, LEDs for simple visual indications, a buzzer for audio indication, counting of objects from 0 to 9999 and user-selectable counting up or counting down options.

Circuit and working

ATmega16 AVR MCU (IC1) is the heart of the programmable object counter whose circuit is shown in Fig. 2. A voltage regulator 7805 (IC2) is used to provide 5V regulated voltage to drive the circuit including the MCU and LCD display. Four pushbutton switches or tactile switches (S2 through S5) are used to set the number to be counted and the up/down count, with a 16x2 LCD to display count values and

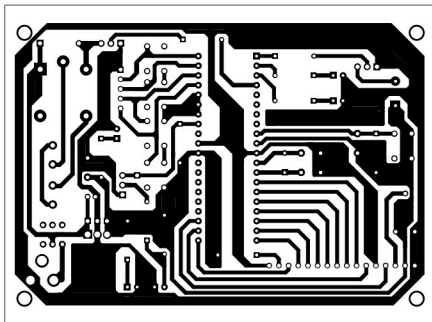


Fig. 3: Actual-size PCB layout of the object counter

other settings.

An object-sensor circuit consists of an infrared (IR) LED (acting as the transmitter) and a photodiode (acting as the receiver) placed 7cm apart, facing each other. The IR beam from the transmitter falls continuously on the receiver. Whenever the IR beam is interrupted, that is, when an object comes in between IRTX1 and IRRX1, a count is registered. In this way, the pair of IR transmitter (IRTX1) and IR receiver (IRRX1) continuously monitors the objects.

The motor to control the conveyor belt is connected across relay RL1. Transistor T2 is the relay driver connected to port pin PB3 of ATmega16 (IC1).

Port pin PB0 of IC1 drives the DC buzzer through transistor T1. Green LED1 and red LED2 are connected to pins PB1 and PB2 to indicate count-progress and count-stop, respectively. An 8MHz crystal is connected to crystal input pins of IC1 to provide an internal clock for the operation.

IRTX1 and IRRX1 are given 5V supply through current-limiting resistors R9 and R8, respectively. Photodiode IRRX1 is connected in reverse mode through R8. It is also connected to the base of transistor

T3 through R5. The collector of T3 is connected to port pin PA7 of IC1. The IR beam interruption by the object is reflected at the collector of transistor T3. The interruption is monitored by the program in the microcontroller (IC1).

When the desired count is reached, the relay is switched off and hence, the motor pulling the conveyor belt stops.

Software program

Circuit operation is done using the software program loaded into the internal memory of the AVR MCU. The program implements all required functionalities including handling user inputs through pushbuttons, displaying messages on the LCD, switching on and switching off the load connected through the relay and visual/audio indications.

The program is written in C language. AVR studio software tool is used to compile the program.

Construction and testing

An actual-size, single-side PCB layout of the circuit is shown in Fig. 3 and its component layout in Fig. 4. After soldering the components properly, a 12V DC adaptor is connected to the